

Observable Dynamics of Sectorized Observable Frameworks

Typed Deformations, Observable Trajectories, and Wall Pullbacks

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2026

This paper (Paper IX of the RIME program) develops the typed deformation layer over Paper VIII's static SOF object language while keeping trajectories, wall pullbacks, and response diagnostics relative to declared charts, paths, carriers, and policies.

Abstract

Problem. Paper VIII defines a SOF together with a marked sectorization, a labelled observable alphabet, and separate operator/word and Lie/Hall carriers. This paper asks how these typed static objects can be compared across a declared deformation family without conflating carrier changes, parameterization, or finite-cutoff observations.

Approach. We develop the corresponding dynamic layer. On an admissible chart, a deformation has the form

$$\mathcal{F}_t = (V_t, Q(t), Y(t); X(t), \mathcal{H}), \quad t \in U \subseteq T,$$

where $\mathcal{V} \rightarrow U$ is a finite-rank Hermitian bundle or a declared local trivialization, the label sets and filtration conventions are fixed, and each field is included only when that carrier is present. A typed comparison map places the selected fibrewise data into a fixed target space. Observable trajectories are obtained only after choosing a one-parameter path in the chart. The dynamic observables are therefore typed fields such as

$$R_1[Y](t), \quad \text{Route}_d[Y](t), \quad W_d[Y](t), \quad D_{\text{word}}^{(\leq d_{\max})}(t),$$

and, independently,

$$R_1^{\text{Lie}}(t), \quad R_2^{\text{Lie}}(t), \quad D_{\text{Lie}}^{(\leq d_{\max})}(t).$$

Results. The paper proves the Observable Wall Pullback Theorem: under explicit admissibility and local-constancy hypotheses, the wall of a discrete shadow is contained in the pullback of the corresponding typed rank, support, collision, or filtration discriminant. Equality requires pullback-exactness on the selected chart; a transverse one-parameter crossing supplies a geometric sufficient condition. An exact three-sector construction with equally normalized generators separates first-order direct norm scaling from second-order simple-commutator norm scaling under a declared threshold policy. A second exact exponential model proves policy-relative half-response ordering under a declared calibration, and a finite three-sector realization certifies $30 < 1380$. It also records an endpoint-normalized, computationally observed hierarchy of continuous neural-network proxies, $\tau_{50}^{\text{end}}(K_0) < \tau_{50}^{\text{end}}(K_1) < \tau_{50}^{\text{end}}(K_2)$, while explicitly leaving the proxy-to-discrete-shadow bridge open.

Boundary. The scope is deliberately narrow: the typed static language is shared, but observable dynamics and wall geometry remain species- and deformation-dependent.

Notation Table

Symbol	Meaning
$\mathcal{F}$	a declared SOF object from Paper VIII
$\mathcal{F}_t$	SOF at deformation parameter t
T	deformation parameter space
$\mathcal{V} \rightarrow U$	finite-rank Hermitian bundle on one typed deformation chart
V_t	fibre of $\mathcal{V}$ at t
I, A, G_0	fixed sector, operative-alphabet, and Lie label sets on a chart
$Q(t)$	declared sector projectors, if the sectorization moves
$Y(t)$	labelled operator alphabet, if the operator branch is present
$X(t)$	registered Lie generator family, if the Lie branch is present
$\mathcal{H} = \{\mathcal{H}_d\}_{d \geq 0}$	fixed Lie/Hall formal-expression filtration and depth convention on a chart
Θ_κ	comparison/extraction map into the fixed target $\mathcal{E}_\kappa$
SOF_{def}	interface for declared deformation records
$A_Y^+(t)$	positive associative word algebra generated by $Y(t)$
$A_Y^*(t)$	observable star-closure generated by $Y(t)$
$A_{Q,Y}^*(t)$	sector-enriched star-closure generated by $D_{Q(t)} \cup Y(t)$
$\mathcal{R}_{d,ij}[Y](t)$	routed-product space from sector j to sector i
$\mathcal{W}_{d,ij}[Y](t)$	full-word corner space from sector j to sector i
$R_1[Y]$	direct support for a labelled operator family
$\text{Route}_d[Y]$	routed-product shadow at route depth d
$W_d[Y]$	full labelled word shadow at word length d
$D_{\text{route}}[Y], D_{\text{word}}[Y]$	exact first-hit depths in $\mathbb{N} \cup \{\infty\}$
$D_\kappa^{(\leq d_{\max})}$	truncated depth in $\{1, \dots, d_{\max}, \text{unreached}\}$
R_1^{Lie}	direct support for a registered Lie family
R_2^{Lie}	projected simple-commutator support
D_{Lie}	exact first-hit depth in a declared Hall/Lie filtration
J_κ	continuous target-space field for typed carrier κ
$\hat{O}_\kappa$	target-space feature map evaluated on J_κ
Σ_O	wall of a selected shadow O
Δ_κ	discriminant in the target space of J_κ
$\gamma : I_\gamma \rightarrow U$	selected one-parameter trajectory through a chart
$\tau_\eta(O_{\kappa,\gamma})$	threshold time relative to γ , normalization, norm, and policy

The symbols R_1 , R_2 , and D without a carrier qualifier are not used as dynamic objects below. Exact and truncated depths use different symbols. The value `unreached` belongs only to a truncated field and is not mathematical infinity.

1 Introduction

Paper VIII answers the static question: what data constitute a SOF and which typed constructions are functorial under strict morphisms [5]? This paper asks the next question:

How do selected SOF observables change along a declared deformation?

The answer cannot be a universal ladder. A deformation may move projectors, move labelled operators, move both, or change only a concrete embedding while leaving an abstract algebra type unchanged. Even for a single alphabet, the positive word algebra A_Y^+ , the observable star-closure A_Y^* , and the sector-enriched star-closure $A_{Q,Y}^*$ are distinct. Each closure choice produces a different observable field. The common object is the pipeline for constructing and measuring them, not a universal law of dynamics.

Schematically, the typed static data vary as

$$t \mapsto (Q(t), Y(t), E_Y(t), \{\mathcal{R}_d[Y](t), \mathcal{W}_d[Y](t)\}_{d \geq 1}, A_Y^+(t), A_Y^*(t), A_{Q,Y}^*(t), X(t), \mathcal{H}),$$

with only the components present in the declared realization retained. This single notation does not merge their wall types. It instead separates direct support walls, routed-space rank walls, word-space dimension walls, star-algebra type walls, word-depth walls, and Lie-depth walls. Closure fields record what is ultimately generated; finite filtrations record how and at what first length it is generated.

The central distinction is:

Sectorization is source-dependent, while comparison data are chart-dependent. The typed object and deformation-record interfaces are shared; observable dynamics remain deformation-dependent.

The distinction is operational. A deformation family over T is not yet a time trajectory. A trajectory additionally chooses a map $\gamma : I_\gamma \rightarrow T$ from an ordered real interval and a typed chart along its image. A wall is not a primitive object of the static SOF. It is a failure of local constancy of a selected shadow on a chart or along such a trajectory.

1.1 Dynamic Data Are Typed

For an operator SOF, the available fields may include

$$R_1[Y](t), \quad \text{Route}_d[Y](t), \quad W_d[Y](t), \quad D_{\text{route}}^{(\leq d_{\max})}[Y](t), \quad D_{\text{word}}^{(\leq d_{\max})}[Y](t).$$

For a separately registered Lie/Hall carrier, the available fields may include

$$R_1^{\text{Lie}}(t), \quad R_2^{\text{Lie}}(t), \quad D_{\text{Lie}}^{(\leq d_{\max})}(t).$$

These two lines represent parallel audit branches. No equality between word support, routed support, commutator support, and Lie depth is assumed. A promotion from one line to another needs a separate theorem or certificate.

The Boolean routed and word shadows sit over the actual moving corner spaces

$$\mathcal{R}_{d,ij}[Y](t) \quad \text{and} \quad \mathcal{W}_{d,ij}[Y](t).$$

Their dimensions and ranks can define walls even when a thresholded support shadow does not change. They must also be distinguished from the saturated corners

$$Q_i(t)A_Y^+(t)Q_j(t), \quad Q_i(t)A_Y^*(t)Q_j(t), \quad Q_i(t)A_{Q,Y}^*(t)Q_j(t).$$

The first is positive-word saturated, the second is star-word saturated, and the third permits sector projectors as internal route separators.

1.2 Exact and Truncated Depth Fields

For any declared route, word, or Hall filtration with level shadows $O_{\kappa,d}(t)$, the exact first-hit depth is

$$D_\kappa(t) = \inf\{d \geq 1 : O_{\kappa,d}(t) = 1\} \in \mathbb{N} \cup \{\infty\}.$$

The corresponding cutoff record is a different field:

$$D_\kappa^{(\leq d_{\max})}(t) = \begin{cases} D_\kappa(t), & D_\kappa(t) \leq d_{\max}, \\ \text{unreached}, & \text{otherwise.} \end{cases}$$

An exact finite first-hit depth $D_\kappa(t) = d$ may be reported only from a first-hit certificate consisting of a witness at level d together with verified non-hits at every lower level $1, \dots, d-1$. A witness at level d alone establishes only

$$D_\kappa(t) \leq d.$$

Exact infinity additionally requires an appropriate closure or saturation certificate. A finite computation must therefore distinguish: first hit certified at d ; hit observed by d with minimality unaudited; and unreached through $d_{\max}$. The last case has truncated-field value unreached and compiler-facing result state `UNREACHED_AT_CUTOFF`; neither denotes mathematical infinity. Consequently, a cutoff wall $\Sigma_{D_\kappa^{(\leq d_{\max})}}$ may depend on $d_{\max}$ and must not be identified with an exact-depth wall.

1.3 Deformation Classes

The following cases are distinct:

Class	Moving data	Typical question
fixed sectors	$Q(t) = Q$, $Y(t)$ varies	how does one family redistribute support?
moving sectors	$Q(t)$ varies, $Y(t) = Y$	how does a changing coarse coordinate system alter shadows?
joint deformation	both $Q(t)$ and $Y(t)$ vary	which field is responsible for a wall?
positive word closure	$A_Y^+(t)$ varies	does positive-word saturation change?
observable	$A_Y^*(t)$ varies	does adjoint closure change the corner structure?
star-closure		
sector-enriched	$A_{Q,Y}^*(t)$ varies	do marked routed corners change?
closure		
abstract star-algebra	a Wedderburn type of A_Y^* or $A_{Q,Y}^*$ is compared	what is invisible to abstract type alone?

An unchanged abstract Wedderburn type of either star-closure does not imply that the marked sector corners remain unchanged; a changed concrete embedding does not by itself imply a change in the abstract star-algebra type. No Wedderburn claim is made for A_Y^+ unless semisimplicity is established separately.

2 Related Work and Novelty Boundary

The external results below provide precedents for dynamic phenomena. They are not theorems about SOF.

2.1 Parameter-Space Rate Separation

Xu, Vardi, and Safran prove a rate separation in ridge regression between a data-visible row-space component and an orthogonal component controlled only by weight decay [7]. Writing

$$\theta = \theta_{\parallel} + \theta_{\perp},$$

their result gives a fast channel driven by the empirical loss and a slow channel driven by regularization. This provides a mechanism-separated precedent for asking whether a comparable separation can be measured in observable space, without identifying $\theta_{\parallel}$ or $\theta_{\perp}$ with any SOF carrier.

2.2 Long Plateaus

Prethermalization provides a precedent for fast transients followed by long plateaus and delayed visibility of slower effects [1]. This motivates studying plateau observables in SOFs, but no prethermalization law is assumed here.

2.3 Wall-Crossing Formulae

Kontsevich and Soibelman provide settings in which crossing a wall is accompanied by an exact transformation law [6]. The theorem below proves only a pullback inclusion for selected observable walls. An SOF wall-crossing transformation law remains open.

The static typed SOF object language is inherited from Paper VIII [5]. Papers V and VII supply independent low-order channel and incidence interfaces [2, 3]; Paper VI supplies normality-gated pointwise registrations and a linearized research interface [4]. These papers are compatible interfaces, not a linear proof chain.

The contribution here is the typed deformation layer: admissible deformation charts, response observables along declared trajectories, typed wall pullbacks, and rate constructions under declared policies. It does not promote continuous proxies to binary shadows, identify distinct carriers, or supply a universal wall-crossing law.

3 Typed SOF Deformations

3.1 Deformation Families

Let

$$\mathcal{F} = (V, Q, Y; X, \mathcal{H})$$

denote a typed SOF record. The entries after the semicolon are optional enrichments; the operator alphabet Y and the Lie family X are not silently identified.

A deformation over a parameter space T is a family of typed SOF records. The notation

$$\mathcal{F}_t = (V_t, Q(t), Y(t); X(t), \mathcal{H}), \quad t \in T,$$

is meaningful pointwise, but it does not itself compare different fibres. Continuity, walls, and response times are defined only after the required local comparison data have been supplied.

3.2 Typed Deformation Charts

Definition 3.1 (Typed Deformation Chart).

Let $U \subseteq T$. A κ -typed deformation chart on U consists of:

1. a finite-rank Hermitian vector bundle $\pi : \mathcal{V} \rightarrow U$ with fibre V_t , together with a local trivialization whenever matrix coordinates are used;
2. fixed finite label sets I , A , and, when present, G_0 for sectors, operative letters, and Lie generators;
3. continuous sections $Q_i(t), Y_a(t), X_g(t) \in \text{End}(V_t)$ for the declared labels, where the $Q_i(t)$ form a complete orthogonal sectorization;
4. a fixed operative word convention, positive or explicitly star-completed, and a fixed Lie/Hall formal-expression filtration $\mathcal{H} = \{\mathcal{H}_d\}_{d \geq 0}$ and depth convention on U ;
5. a fixed target space $\mathcal{E}_\kappa$ and a declared continuous comparison/extraction map Θ_κ from the selected fibrewise carrier data into $\mathcal{E}_\kappa$.

The resulting target-space field is

$$J_\kappa(t) = \Theta_\kappa(V_t, \{Q_i(t)\}, \{Y_a(t)\}; \{X_g(t)\}, \mathcal{H}).$$

Only the components required by κ are retained. A globally fixed finite-dimensional space is the special case of a trivial bundle.

Because a continuous finite-dimensional projector family has locally constant rank, a sector-rank change cannot occur inside one connected typed chart. Changes in sector rank, label sets, operative alphabets, word conventions, or Hall conventions are **chart-transition or schema events**. They are not ordinary walls of a fixed typed field unless an additional comparison construction is declared.

3.3 The Deformation Record Interface

The notation SOF_{def} denotes the class of deformation records used here. A record from $\mathcal{F}$ to $\mathcal{G}$

$$\mathcal{F} \rightsquigarrow \mathcal{G}$$

consists of:

1. a parameter space T with marked endpoints t_- and t_+ ;
2. a family $(\mathcal{F}_t)_{t \in T}$ satisfying $\mathcal{F}_{t_-} = \mathcal{F}$ and $\mathcal{F}_{t_+} = \mathcal{G}$ after the declared endpoint identifications;
3. a typed deformation chart covering each domain on which a selected field is compared;
4. endpoint comparison maps, and chart-transition data when the endpoint fibres or chart schema differ;
5. a cutoff, threshold, and saturation convention where relevant.

These deformation records are not declared morphisms in a category. Concatenation would require a specified gluing rule, a reparameterization convention such as Moore paths, and a proof of associativity. None is imposed here. A deformation datum may be non-isometric or may change sector dimensions, but it must state which observables it preserves or tracks. The notation keeps the static strict category of Paper VIII separate from this declared dynamic comparison interface.

3.4 Admissibility

For a selected carrier κ , a deformation is κ -admissible on $U \subseteq T$ when:

1. a κ -typed deformation chart is declared on U ;
2. its comparison map defines a continuous field $J_\kappa : U \rightarrow \mathcal{E}_\kappa$;
3. the threshold, cutoff, and saturation policy are fixed on U ;
4. the target-space feature map is locally constant away from its declared discriminant;
5. any exact-depth claim states the closure or saturation certificate that distinguishes ∞ from cutoff-unreached.

Admissibility is diagnostic-specific: a family can be admissible for a continuous norm proxy yet inadmissible for a binary depth field.

A deformation of the Hall convention itself is outside the default chart definition. Such a filtration-schema deformation requires level-identification maps between the changing conventions and is registered as a schema event, not silently treated as a value change of D_{Lie} .

3.5 Carrier Packages

The compact notation

$$J_{\text{acc}} = (J_{\text{op}}, J_{\text{route}}, J_{\text{word}}, J_{\text{Lie}})$$

is allowed only as an explicitly enumerated package. The individual fields have different meanings:

Carrier	Field examples	Typical discriminant
operator	$B_{ij}^a(t), R_1[Y](t)$	block support threshold
routed	$\mathcal{R}_{d,ij}[Y](t), \text{Route}_d[Y](t)$	routed-product-space rank/support
word	$\mathcal{W}_{d,ij}[Y](t), W_d[Y](t),$ $D_{\text{word}}^{(\leq d_{\text{max}})}(t)$	word-space rank or truncated first hit
positive associative closure	$A_Y^+(t)$	concrete algebra, corner, or dimension change; no automatic Wedderburn type
star closures	$A_Y^*(t), A_{Q,Y}^*(t)$	concrete corner or finite-dimensional C^* -algebra Wedderburn-type change
Lie/Hall	$X_g(t), R_2^{\text{Lie}}(t), D_{\text{Lie}}^{(\leq d_{\text{max}})}(t)$	commutator rank/support or truncated Hall first hit

The table is an audit order, not an implication diagram.

4 Observable Trajectories

Let a deformation family be declared over T , let $U \subseteq T$ carry a κ -typed deformation chart, and choose a continuous one-parameter path

$$\gamma : I_\gamma \longrightarrow U, \quad I_\gamma \subseteq \mathbb{R}.$$

The associated **observable trajectory** is

$$O_{\kappa,\gamma}(s) = \widehat{O}_\kappa(J_\kappa(\gamma(s))), \quad s \in I_\gamma.$$

Thus, although a deformation family can be studied over a general topological parameter space, a response time is attached only to a selected ordered path. Different paths through the same family need not define the same trajectory.

Examples include:

$$\begin{aligned} &R_1[Y](t), \text{Route}_d[Y](t), W_d[Y](t), \\ &D_{\text{route}}^{(\leq d_{\max})}[Y](t), D_{\text{word}}^{(\leq d_{\max})}[Y](t), \\ &R_1^{\text{Lie}}(t), R_2^{\text{Lie}}(t), D_{\text{Lie}}^{(\leq d_{\max})}(t), \\ &P_d^{\text{state}}(t), \text{gap}(t), \text{proxy}(t). \end{aligned}$$

The last line contains separately registered state, spectral, or proxy observables, not automatic identifications with the operator or Lie branches. In particular, P_d^{state} requires its own state filtration and depth convention; it is not the unqualified plateau of D_{word} or D_{Lie} .

Typed Observable Trajectory

A typed chart makes fibres comparable; a selected path then defines a trajectory.

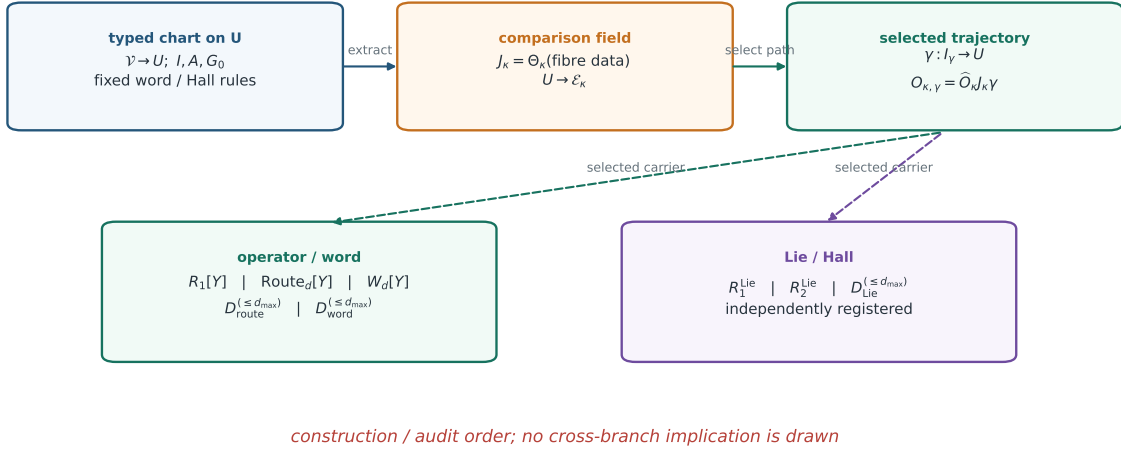


Figure 1. Typed observable trajectory. A typed chart first makes the selected fibrewise data comparable in one target space; a one-parameter path then defines a trajectory for one carrier. Different branches are not merged by the diagram.

4.1 Response Times

Let $O_{\kappa, \gamma} : I_\gamma \rightarrow \mathbb{R}$ be a continuous scalar observable on an ordered interval with initial point s_0 . For a declared threshold η , define the first-crossing parameter

$$\tau_\eta(O_{\kappa, \gamma}) = \inf\{s \in I_\gamma : s \geq s_0, O_{\kappa, \gamma}(s) \geq \eta\}.$$

The extended-real convention is $\inf \emptyset = +\infty$. A finite sampled or bounded observation record must instead carry a declared right-censoring record, which may be displayed as **UNREACHED_ON_DECLARED_INTERVAL**; it cannot promote non-crossing on that interval to an intrinsic infinite response time. **UNREACHED_ON_DECLARED_INTERVAL** is a response-measurement censoring value, not the filtration result state **UNREACHED_AT_CUTOFF**. Under Compiler v1.0, it is represented by an **OBSERVED** response-time finding with null numerical value and a referenced sampling or trajectory policy declaring right censoring; it is not a new generic result state.

Other response conventions, such as half-response or interval-censored wall time, must be declared separately. Response times are defined only relative to the chosen trajectory parameterization, observable normalization, norm, orientation, and threshold policy. A nonlinear reparameterization of γ or a rescaling of the observable generally changes τ_η .

For discrete shadows, a wall time may be interval-censored:

$$\tau_{\text{wall}}(O_{\kappa,\gamma}) \in [s_m, s_{m+1}]$$

when the sampling grid identifies only the first interval containing the change. A proxy half-response is not a discrete wall time.

5 Typed Observable Walls

5.1 Wall Definition

For a discrete shadow O_κ on a κ -admissible domain U , define

$$\Sigma_{O_\kappa} = \{t \in U : O_\kappa \text{ is not locally constant at } t\}.$$

For a selected trajectory $\gamma : I_\gamma \rightarrow U$, define analogously

$$\Sigma_{O_{\kappa,\gamma}} = \{s \in I_\gamma : O_{\kappa,\gamma} \text{ is not locally constant at } s\}.$$

For a continuous field, the corresponding object is a declared discriminant where rank, support, collision, filtration type, or another qualitative regime changes. Both the continuous field and its threshold policy must be named explicitly.

Examples are

$$\Sigma_{R_1[Y]}, \quad \Sigma_{\text{Route}_d[Y]}, \quad \Sigma_{W_d[Y]}, \quad \Sigma_{R_1^{\text{Lie}}}, \quad \Sigma_{R_2^{\text{Lie}}}, \quad \Sigma_{D_{\text{Lie}}^{(\leq d_{\max})}}.$$

The notation Σ_{access} is permitted only when it is defined as a named union of these typed walls with a fixed domain and policy. It is not a primitive universal wall.

5.2 Observable Wall Pullback Theorem

Theorem 5.1 (Observable Wall Pullback Inclusion).

Let $U \subseteq T$ carry a κ -typed deformation chart and be κ -admissible. Let $\mathcal{E}_\kappa$ be a locally connected topological space, and let

$$J_\kappa : U \longrightarrow \mathcal{E}_\kappa$$

be continuous. Let $\Delta_\kappa \subseteq \mathcal{E}_\kappa$ be closed, and suppose the target feature map $\widehat{O}_\kappa$ is constant on every connected component of $\mathcal{E}_\kappa \setminus \Delta_\kappa$. For $O_\kappa = \widehat{O}_\kappa \circ J_\kappa$, one has

$$\Sigma_{O_\kappa} \subseteq J_\kappa^{-1}(\Delta_\kappa).$$

Proof. Because Δ_κ is closed, its complement is open. Local connectedness implies that each connected component of this complement is open. If $t \notin J_\kappa^{-1}(\Delta_\kappa)$, continuity of J_κ therefore guarantees a neighborhood of t mapped entirely into one such component. The observable $O_\kappa = \widehat{O}_\kappa \circ J_\kappa$ is constant on that neighborhood, which proves the inclusion.

Definition 5.2 (Pullback-Exact Discriminant).

The discriminant Δ_κ is pullback-exact for $(J_\kappa, \widehat{O}_\kappa)$ on U when, for every $t \in J_\kappa^{-1}(\Delta_\kappa)$ and every neighborhood N of t in U , there exist $t_1, t_2 \in N$ such that

$$\widehat{O}_\kappa(J_\kappa(t_1)) \neq \widehat{O}_\kappa(J_\kappa(t_2)).$$

Corollary 5.3 (Equality under Pullback-Exactness).

Under the hypotheses of Theorem 5.1, if Δ_κ is pullback-exact for the selected chart, then

$$\Sigma_{O_\kappa} = J_\kappa^{-1}(\Delta_\kappa).$$

Proof. Theorem 5.1 gives one inclusion. Pullback-exactness makes every point of the pullback a failure of local constancy, giving the reverse inclusion.

An ambient discriminant can fail to be pullback-exact when the deformation is constant inside it, tangent to it, or confined to one feature stratum. Thus ambient exactness alone does not imply equality. The theorem is a pullback statement for one selected carrier; it does not identify different carriers or prove that a typed wall exists for every deformation.

Proposition 5.4 (Transverse Trajectory Crossing).

Suppose $\mathcal{E}_\kappa$ is a smooth manifold and, near $J_\kappa(\gamma(s_0))$, the discriminant Δ_κ is a smooth embedded hypersurface separating two local components on which $\widehat{O}_\kappa$ takes different values. Assume s_0 is an interior point of I_γ , and that $J_\kappa \circ \gamma$ is C^1 on an open interval containing s_0 , with $J_\kappa(\gamma(s_0)) \in \Delta_\kappa$. If

$$\frac{d}{ds}(J_\kappa \circ \gamma)(s_0) \notin T_{J_\kappa(\gamma(s_0))}\Delta_\kappa,$$

then $J_\kappa \circ \gamma$ crosses Δ_κ transversely at s_0 and

$$s_0 \in \Sigma_{O_{\kappa,\gamma}}.$$

Proof. Transversality to a locally separating hypersurface places the trajectory on opposite local sides of the discriminant for parameters arbitrarily close to s_0 . The feature values on those sides differ, so $O_{\kappa,\gamma}$ is not locally constant at s_0 .

Call Δ_κ **trajectory-exact** for $J_\kappa \circ \gamma$ when every

$$s \in (J_\kappa \circ \gamma)^{-1}(\Delta_\kappa)$$

is a local nonconstancy point of $O_{\kappa,\gamma}$. Consequently, a one-parameter trajectory for which every discriminant encounter satisfies Proposition 5.4 is trajectory-exact for $J_\kappa \circ \gamma$, equivalently pullback-exact after restriction to that trajectory. Tangencies, motion inside the discriminant, and crossings between sides carrying the same feature value remain outside this sufficient condition.

5.3 Paper VI Boundary

Paper VI supplies a normality-gated interface for pointwise registrations and linearized computational certificates. In the notation used here, its spectral gates are relevant only on domains where the selected spectral construction is defined:

$$\Sigma_{\text{comm}} \supseteq \Sigma_{\text{normal}} \supseteq \Sigma_{\text{spec}}^{(\nu)}.$$

Those registrations may serve as inputs to a moving-field theory. They are not used here as a positive moving SOF instance, and the nonnormal fragmentation calculations do not support a universal wall hierarchy.

Typed Wall Pullback

A chart field pulls a target discriminant back; a path crossing is a separate test.

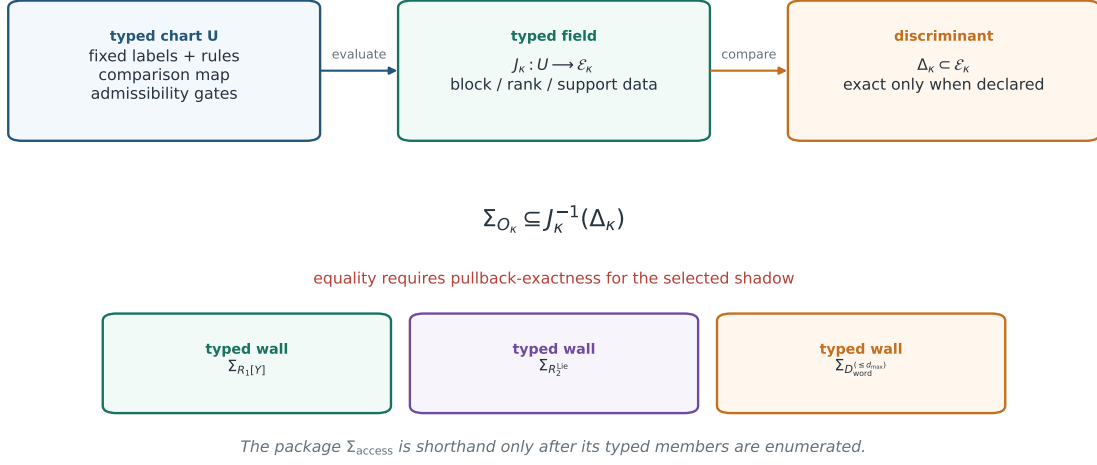


Figure 2. Typed wall pullback. A chart field pulls a target discriminant back to the deformation domain. Equality requires pullback-exactness; transverse path crossing is a separate sufficient test.

6 Deformation Species and Boundary Cases

6.1 Operator and Generator Deformations

For a fixed sectorization, one may vary the labelled operator family:

$$Y_a \mapsto Y_a(w).$$

The resulting fields are $R_1[Y](w)$, routed shadows, or word shadows only after the route, word, threshold, and cutoff conventions are fixed. The moving closure must be identified explicitly as $A_Y^+(w)$, $A_Y^*(w)$, or $A_{Q,Y}^*(w)$. For the two star-closures, the concrete embedding or marked corners can change without a change in abstract Wedderburn type. The positive algebra $A_Y^+(w)$ carries no such type unless semisimplicity is separately proved. No generic fragmentation, plateau, or oscillation law is claimed.

6.2 State-Mixing Deformation

A Yang-like state-mixing family may be written

$$\rho(\varepsilon) = (1 - \varepsilon)\rho_0 + \varepsilon\sigma.$$

The associated plateau or mixing observable belongs to the registered state carrier. It need not equal an operator accessibility field. Here the family serves only to distinguish state deformation from operator deformation. No typed wall or rate law is inferred from the available plateau summaries.

6.3 Markov, Graph, and Quantum Deformations

Markov, graph, and quantum systems may vary transition operators, adjacency or Laplacian data, Hamiltonians, or gate families. Their registered shadows can include communicating-class profiles, connectedness, spectral gaps, or word/Lie accessibility. A discrete rewiring or an

unstructured interpolation does not automatically define a smooth rate hierarchy. The carrier and deformation law must be declared before a wall or response time is compared.

6.4 Training-Coupled Neural SOF

In the neural diagnostic listed as Appendix A (A3), activation-induced sectors and trainable weight operators define a small SOF-like realization. The measured quantities are continuous block-norm proxies:

$$K_0(t), \quad K_1(t), \quad K_2(t),$$

for direct blocks, simple-commutator blocks, and nested-commutator blocks. These continuous proxies are not the typed binary fields R_1^{Lie} , R_2^{Lie} , or $D_{\text{Lie}}^{(\leq d_{\text{max}})}$.

The audit uses the final declared sample t_{end} , not an asymptotic limit. For each nonflat proxy, it defines the endpoint-normalized displacement

$$\widehat{K}_r^{\text{end}}(t) = \frac{K_r(t) - K_r(0)}{K_r(t_{\text{end}}) - K_r(0)}$$

and reports the first sampled half-response

$$\tau_{50}^{\text{end}}(K_r) = \min \left\{ t \in \mathcal{G} : \widehat{K}_r^{\text{end}}(t) \geq \frac{1}{2} \right\},$$

where $\mathcal{G}$ is the declared training-audit grid. Thus these quantities are normalized by the declared endpoint and use a sampled half-response policy. They are distinct from the asymptotic normalization introduced in the calibrated construction below.

The default run gives

$$\tau_{50}^{\text{end}}(K_0) = 60, \quad \tau_{50}^{\text{end}}(K_1) = 80, \quad \tau_{50}^{\text{end}}(K_2) = 120.$$

Activation	$\tau_{50}^{\text{end}}(K_0)$	$\tau_{50}^{\text{end}}(K_1)$	$\tau_{50}^{\text{end}}(K_2)$	pointwise cutoff audit
ReLU	60	80	120	(0, 0, 0)
GeLU	60	80	120	(0, 0, 0)

This is a computational observation of ordered proxy response times. The pointwise triples report, in order, unsupported direct pairs, Lie-emergent pairs, and cutoff-unreached Lie pairs. They do not define a temporal repair event: the diagnostic does not coherently continue sector labels across training time. In particular, it does not measure $\tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$.

6.5 Proxy Shadow Boundary

The neural diagnostic exposes two layers:

$$(K_0, K_1, K_2) \quad \text{and} \quad (R_1^{\text{Lie}}, R_2^{\text{Lie}}, D_{\text{Lie}}^{(\leq d_{\text{max}})}).$$

No theorem identifies these layers. In particular,

$$K_2(t) \not\Rightarrow D_{\text{Lie}}^{(\leq d_{\text{max}})}(t)$$

without a threshold, margin, sector-stability, and filtration-compatibility result. We call the missing result the **Observable Proxy Shadow Principle**. It remains a Research Program rather than an assumption.

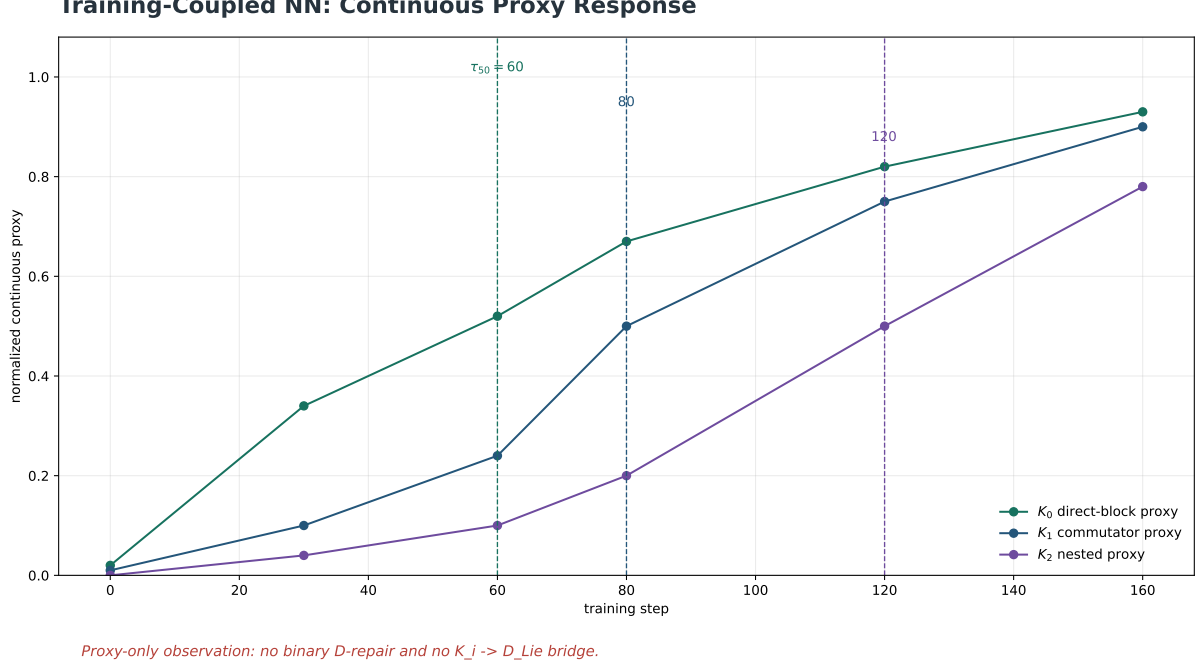


Figure 3. Training-coupled neural proxies. The ordered response times are shown as continuous proxy evidence only; no bar is labeled as a binary depth event.

7 Diagnostic Contrast and Rate Separation

7.1 Species Dependence

State mixing changes a state carrier, while an operator deformation changes a labelled alphabet or its concrete embedding. Similar scalar summaries do not identify the underlying typed fields. Accordingly, the Rubik generator-weight fragmentation/oscillation diagnostic is excluded from the typed evidence and from the present comparison.

7.2 Observable Rate Separation

For selected observables $O_1, \dots, O_k$ on the same parameterized trajectory γ , with fixed normalization, norm, orientation, and response policy, the trajectory is called observable-rate-separated when at least two characteristic times

$$\tau(O_{1,\gamma}), \dots, \tau(O_{k,\gamma})$$

are demonstrably separated. This rate separation is a property of the registered trajectory and policy, not of the underlying static SOF alone.

7.3 Exact Scale-Separated Threshold Construction

Let Q_1, Q_2, Q_3 be the coordinate projectors on $\mathbb{C}^3$, and put

$$J_{ab} = E_{ab} - E_{ba}, \quad X_1(t) = tJ_{12}, \quad X_2(t) = tJ_{23}, \quad t \geq 0.$$

Both declared generators are skew-Hermitian and have the same Frobenius scale at $t = 1$. Along the declared trajectory $\gamma(t) = t$, select the continuous block observables

$$K_{\text{dir}}(t) = \|Q_1 X_1(t) Q_2\|_F, \\ K_{\text{comm}}(t) = \|Q_1 [X_1(t), X_2(t)] Q_3\|_F.$$

Proposition 7.1 (Exact Scale-Separated Threshold Construction).

For the declared generator normalization, trajectory parameterization, Frobenius norm, and common threshold $0 < \eta < 1$, the selected block norms satisfy

$$K_{\text{dir}}(t) = t, \quad K_{\text{comm}}(t) = t^2.$$

Their first-crossing parameters are

$$\tau_{\eta}(K_{\text{dir}}) = \eta, \quad \tau_{\eta}(K_{\text{comm}}) = \sqrt{\eta},$$

respectively.

Proof. The identity $[J_{12}, J_{23}] = J_{13}$ and the declared normalization give the displayed first- and second-order norm scalings. Solving $t = \eta$ and $t^2 = \eta$ gives the two crossing parameters.

Consequently,

$$\frac{\tau_{\eta}(K_{\text{comm}})}{\tau_{\eta}(K_{\text{dir}})} = \eta^{-1/2} > 1.$$

This construction separates the selected continuous direct and simple-commutator block norms because they appear at different polynomial orders along the declared trajectory. It is an existence construction, not an intrinsic rate invariant of the underlying static SOF. Rescaling either observable, changing the threshold, or nonlinearly reparameterizing the path generally changes the ratio. It also does not prove a separation of Boolean support walls or Lie depth. Appendix A (A1) validates the displayed matrices and records zero formula residual at the default threshold $\eta = 10^{-6}$, where the two crossings are 10^{-6} and 10^{-3} .

7.4 Calibrated Exponential Response Construction

A second construction isolates policy-aligned response times without introducing a new carrier. Assume that the finite limit $K(\infty)$ exists and $K(\infty) \neq K(0)$. Define normalized displacement by

$$\widehat{K}(t) = \frac{|K(t) - K(0)|}{|K(\infty) - K(0)|}.$$

Let

$$\widehat{K}_a(t) = 1 - e^{-\gamma t}, \quad \widehat{K}_b(t) = 1 - e^{-\lambda t}, \quad \gamma > \lambda > 0.$$

Proposition 7.2 (Calibrated Two-Channel Response Separation).

Under the declared parameterization, normalized-displacement convention, and common half-response policy,

$$\tau_{1/2}(\widehat{K}_a) = \frac{\log 2}{\gamma} < \frac{\log 2}{\lambda} = \tau_{1/2}(\widehat{K}_b).$$

Proof. For $c > 0$, the equation $1 - e^{-ct} = 1/2$ has the unique solution $t = (\log 2)/c$. Apply this identity to γ and λ , then use $\gamma > \lambda$.

Appendix A (A2) realizes the two responses as normalized sector-to-sector block norms in a fixed three-sector SOF. With

$$\gamma = \frac{\log 2}{30}, \quad \lambda = \frac{\log 2}{1380},$$

the finite audit returns half-response times $30 < 1380$. This has status **Computational Certificate** for the declared block realization. The mechanism labels provide an interpretation;

the ordered calibration supplies the inequality. Neither the proposition nor the finite certificate is an intrinsic rate invariant, a Boolean support wall, or a Lie-depth statement.

The candidate hierarchy

$$\tau(R_1^{\text{Lie}}) < \tau(R_2^{\text{Lie}}) < \tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$$

is a cutoff-relative structured-dynamics hypothesis, not a universal law. A corresponding exact-depth hierarchy would require an exact closure or saturation certificate. The established evidence is deliberately narrower:

1. Proposition 7.1 gives an exact two-level direct/commutator norm separation;
2. Proposition 7.2 and its finite realization give a calibrated two-channel half-response separation under one declared policy;
3. the NN run observes $60 < 80 < 120$ for continuous proxies;
4. state mixing and unstructured interpolations remain distinct deformation interfaces rather than positive hierarchy evidence.

Consequently, no completed $\tau(R_1^{\text{Lie}}) < \tau(R_2^{\text{Lie}}) < \tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$ trajectory is claimed, much less an exact-depth version.

7.5 Parameter-to-Observable Precedent

The ridge-regression result yields

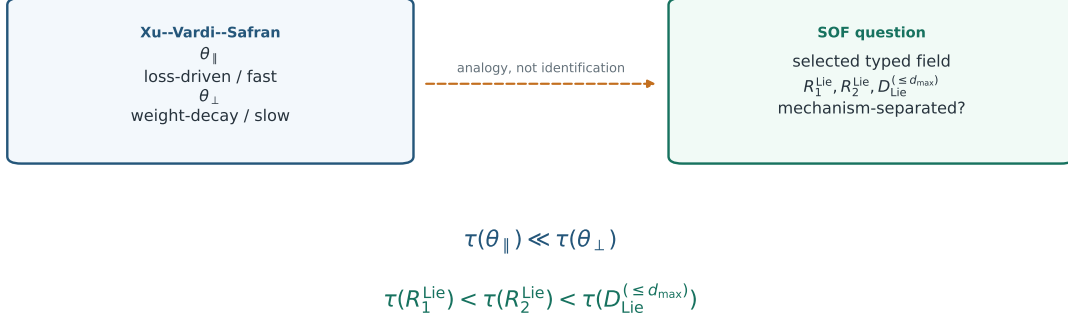
$$\tau(\theta_{\parallel}) \ll \tau(\theta_{\perp}),$$

where the two directions are separated by the training mechanism. The observable-space analogue asks whether a declared mechanism-separated deformation produces ordered response times in a typed carrier. The objects are different, and no map from parameter directions to SOF fields is assumed.

7.6 Dynamic Interface

The role of SOF_{def} is restricted to paths with declared typed charts and comparison data. It is not a category and does not provide a complete weak-morphism theory, a metric on deformation space, or a wall-crossing transformation law.

Parameter Precedent and Observable-Space Question

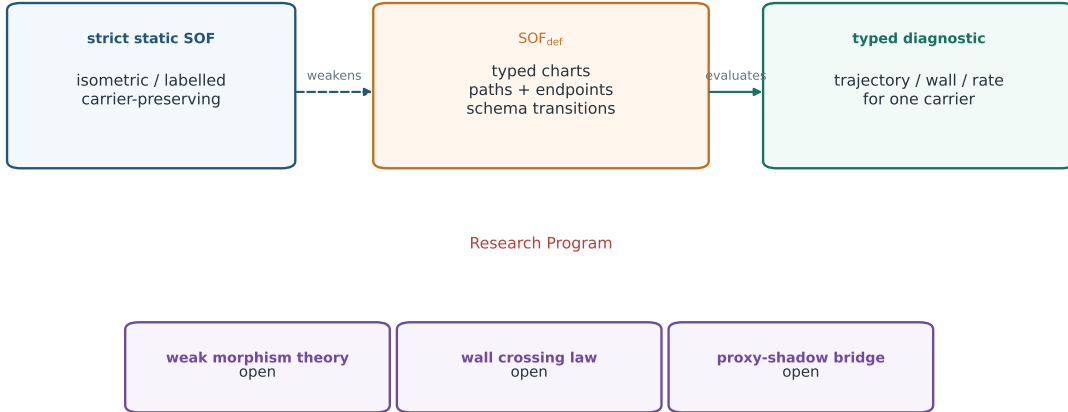


The right-hand hierarchy remains a structured-dynamics target.

Figure 4. Parameter-to-observable comparison. The left side records the theorem-level parameter precedent; the right side records the open SOF question rather than asserting an identification.

SOF Deformation-Record Interface

Typed charts and comparison data support carrier-specific dynamic diagnostics.



This is a record interface, not a completed deformation category.

Figure 5. Deformation-record interface. Typed charts, trajectories, endpoints, and schema transitions support selected diagnostics, but no weak deformation category is asserted.

8 Claim Status and Boundary

Claim	Status
Observable Wall Pullback Inclusion	Theorem
Equality under pullback-exactness	Theorem
Transverse Trajectory Crossing Criterion	Theorem
exact scale-separated threshold construction	Theorem
calibrated two-channel response separation	Theorem
calibrated three-sector block realization with half-response times	Computational Certificate
$30 < 1380$	
NN ordering $\tau_{50}^{\text{end}}(K_0) < \tau_{50}^{\text{end}}(K_1) < \tau_{50}^{\text{end}}(K_2)$	Computational Observation
NN proxy-to-discrete-shadow identification	Research Program
full typed rate hierarchy through truncated or exact D_{Lie}	Research Program
universal wall law or universal deformation geometry	Research Program

The typed deformation record and SOF_{def} define the deformation-record interface; they are not evidence levels. Paper VI's normality-gated pointwise registrations are cited background and do not supply claims here.

8.1 Excluded Claims

The following are not claimed:

1. that one untyped $R_1/R_2/D$ ladder exists for every SOF;
2. that a route, full word, commutator, or Lie depth is determined by another carrier without a bridge theorem;
3. that Paper VI supplies a completed moving-field theorem;
4. that numerical unreached values are infinity;
5. that parameter-space and observable-space rates are identical;
6. that NN proxies exhibit a temporal Lie-depth repair event;
7. that Yang-like state mixing and operator deformation share a wall law;
8. that every deformation is smooth or has one discriminant type;
9. that the full $\tau(R_1^{\text{Lie}}) < \tau(R_2^{\text{Lie}}) < \tau(D_{\text{Lie}}^{(\leq d_{\text{max}})})$ hierarchy has been observed;
10. that response times are invariant under observable rescaling or trajectory reparameterization;
11. that the Observable Proxy Shadow Principle has been proved.

9 Conclusion

The static SOF language extends to declared deformation records without merging operator/word, Lie/Hall, state, or saturated-algebra carriers. The wall-pullback theorem gives a general inclusion and identifies pullback-exactness as the additional condition required for equality; a transverse trajectory crossing supplies a geometric sufficient condition. Proposition 7.1

supplies an exact parameterization- and normalization-relative direct/commutator threshold construction. Proposition 7.2 supplies a calibrated exponential half-response inequality, and its three-sector realization certifies $30 < 1380$ under the declared policy. The NN audit remains continuous proxy evidence only. Together, these results support a typed deformation program rather than a universal wall hierarchy, intrinsic rate invariant, or proxy-to-depth theorem.

10 Outlook

The dynamic evidence supplied to the Paper X compiler follows the construction order

$\text{SOF} \longrightarrow \text{deformation family} \longrightarrow \text{typed observable fields} \longrightarrow \text{wall and rate diagnostics}.$

The arrows denote construction order, not implication.

The next theorem-level targets are:

1. a weak/deformation morphism theory extending the SOF_{def} deformation-record interface;
2. a proxy-to-shadow theorem with margin and threshold hypotheses;
3. a higher-depth structured-dynamics construction with separately controlled direct, commutator, and Lie-depth channels;
4. a genuinely time-resolved $D_{\text{word}}^{(\leq d_{\text{max}})}$ or $D_{\text{Lie}}^{(\leq d_{\text{max}})}$ repair audit, followed where possible by an exact saturation certificate;
5. wall-crossing formulae for restricted species.

These are open extensions, not premises of the results above.

Appendix A Computational Artifacts

The following repository artifacts support the exact response constructions, the finite three-sector certificate, and the neural proxy observations. The default directory is:

`experiments/paper9/`

All short paths below are relative to that directory.

Artifact	Role	Short path
A1	exact direct/commutator threshold construction	<code>rate_hierarchy.py</code>
A2	calibrated exponential block-response realization	<code>calibrated_response.py</code>
A3	training-coupled $K_0/K_1/K_2$ proxy audit	<code>nn_training_sof_tau.py</code>
A4	fixed-weight activation/sectorization audit	<code>nn_activation_sof.py</code>
A5	versioned result and claim-boundary validator	<code>validation/validate_results.py</code>
A6	exact-construction and calibrated-response records	<code>results/rate_hierarchy.json,</code> <code>results/calibrated_response.json</code>
A7	neural proxy and activation records	<code>results/nn_training_sof_tau.json,</code> <code>results/nn_activation_sof.json</code>

From the repository root, run an executable artifact as `python experiments/paper9/<short path>`. A1 checks the exact formulas of Proposition 7.1. A2 supplies the finite Computational Certificate associated with Proposition 7.2. A3 and A4 support only their declared Computational Observations. A5 checks the versioned records and their stated numerical and status invariants.

Archived Rubik generator-weight, plateau, and hard-coded state-mixing scripts are retained under `experiments/paper9/archive/` as historical provenance and do not support the claims above. All listed artifacts are available in the [RIME repository](#).

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